Mao Ding
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Highlight

- Cross-domain data analyst with a hybrid background spanning electrical engineering, biostatistics, and genomics research, bringing rigorous quantitative thinking to large-scale, complex datasets.
- Specialized in statistical genetics and genomics pipeline development, with proven ability to translate complex findings for both technical and non-technical audiences.

Education

Northeastern University — **Portland, ME**
Interdisciplinary PhD — Sep 2025 - Present
Bioinformatics Certificate — Sep 2024 - May 2026
Boston University — **Boston, MA**
Master of Science in Applied Data Analytics (GPA: 3.9/4.0) — Aug 2020 - Aug 2022
Suzhou University of Science and Technology — **Suzhou, China**
Bachelor of Science in Geographic Information System (GIS) — Sep 2002 - Jun 2006

Technical Skills

Programming Languages: Python (Pandas, NumPy, Matplotlib, Scikit-learn), R (dplyr, tidyverse, ggplot2, R Markdown), Java, SQL
Data Science & Machine Learning: ETL Data pipeline, TensorFlow, PyTorch, PySpark, Big Data Analytics, Neural Network, NLP
Statistical Analysis: Regression Model, Hypothesis Testing, Permutation Testing, Statistical Simulation, Bootstrap Resampling
Infrastructure & Tools: Linux/Bash, High-Performance Computing (HPC/SLURM), Git/GitHub, Google Cloud Platform, JMP Pro
Bioinformatics & Genomics: GWAS analysis, PLINK, REGENIE, LD Pruning, genomic file processing (FASTA/Q, BAM, VCF)

Selected Projects

Chromosome Substitution Strain Mouse Genetics of Metabolic Traits

- Conducted comprehensive statistical analysis investigating how genetic background interacts with dietary conditions to influence both mean values and phenotypic variance across metabolic traits in mouse data.
- Applied non-parametric statistic methods, including permutation tests, bootstrap resampling, and multiple-testing correction, to evaluate the additivity of genetic effects across the strain panel.
- Built trait-correlation networks to characterize relationships among metabolic traits, and produced publication-quality figures for a manuscript in preparation.

Parent-of-Origin Effects on Obesity

- Designed end-to-end data analysis pipeline for Framingham Heart Study data, integrating phenotype, genotype, and pedigree information with quality control filtering.
- Ran parental association analyses to identify genetic variants affecting offspring obesity, testing both mean and variance effects.
- Determined parental origin of variant effects by contrasting maternal vs. paternal inheritance patterns in heterozygous individuals using linear regression, Levene's test, and permutation-based significance testing.
- Performed gene expression analyses at single-SNP and haplotype levels to explore the functional mechanisms underlying POE on obesity.

Pharmacogenetic GWAS of the Atenolol for the Prevention of Osteoporosis (APO) Trial

- Conducted a GWAS testing whether SNPs interact with beta-blocker treatment (atenolol vs. placebo) to influence changes in bone traits over time in the APO randomized controlled trial (N=415), identifying variants where treatment response differs by genotype.
- Ran REGENIE and PLINK pipelines on an HPC cluster, identifying 7 genome-wide significant lead SNPs, and modeled genotype-specific treatment-response trajectories for each.
- Prioritized lead variants for follow-up using GTEx, dbSNP, GWAS Catalog, and the UCSC Genome Browser to assess regulatory function and gene targets.

Work Experience

Roux Institute at Northeastern University — **Portland, ME**
Research Data Analyst (Full-time) — Sep 2023 - Present

- Design and maintain end-to-end analytical pipelines integrating phenotype, genotype, and pedigree data, and lead GWAS analyses to translate association signals into biological insights.
- Apply statistical genetics methods to identify variants influencing complex traits - including parent-of-origin effects on obesity and pharmacogenetic treatment response - collaborating cross-functionally to tailor approaches to project-specific needs.

Maine Health Institute for Research — **Portland, ME**
Research Data Analyst (Full-time) — Feb 2023 - Sep 2023

- Worked on NIH-funded research to detect and characterize variance heterogeneity in patient populations, applying statistical methods to identify unusual frequency distributions that standard approaches miss.
- Conducted simulation studies for BifurcatoR, an open-source R toolkit for identifying hidden subgroups in clinical data, comparing statistical power and false positive rates across multiple distributions and detection methods.

State Grid Corporation of China — **China**
Electrical Engineer/Quality and Safety Control Supervisor of Energy Metering Section (Full-time) — Jun 2006 - Jul 2017

- Led and coordinated a team of staff members to maintain and collect smart meter data for over 2 million customers.
- Directed line loss data analysis across 2,000+ low-voltage power supply areas and 52 electrical substations.